AR-RAFI ISHRAQ

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ACADEMIC PROFILE

Final-semester Electrical and Electronic Engineering student at the **Islamic University of Technology (IUT)** with a **CGPA of 3.94/4.00**. Research experience in photonic crystal fiber-based surface plasmon resonance biosensors, finite-element modelling, and sensitivity optimization. Academic and project experience spans photonics, electronic and digital systems, VLSI, embedded systems, signal processing, communication engineering, and power systems. Seeking a Lecturer position in Electrical and Electronic Engineering with strong interest in undergraduate teaching, laboratory instruction, and research.

EDUCATION

Islamic University of Technology (IUT) — Aug 2022 – Expected Sep 2026
B.Sc. in Electrical and Electronic Engineering — **CGPA: 3.94/4.00**
Current Academic Standing: **5th among 164 students**

Notre Dame College — 2019 – 2022
Higher Secondary Certificate — GPA: 5.00/5.00

Ideal School and College — 2017 – 2019
Secondary School Certificate — GPA: 5.00/5.00

HIGHLIGHTED COURSEWORK

Communication: Communication Engineering I & II, Data Communication and Networking I & II, Wireless Communication, Microwave Engineering, Cellular Communication

Signal Processing & Control: Signals and Systems, Digital Signal Processing, Random Signal Processing, Control Systems

Electronics & VLSI: Electronics I & II, Digital Electronics, Semiconductor Physics, Power Electronics, VLSI Design

Embedded Systems: 8051 Microcontroller-Based System Design, 8086 Microprocessor

Power Systems: Power System I, II & III, Power Generation

SCHOLARSHIPS

- Awarded an **OIC Partial Scholarship** upon admission to Islamic University of Technology (IUT).
- Awarded a **General Merit Scholarship** based on performance in the Higher Secondary Certificate (HSC) examination.

RESEARCH INTERESTS

Photonics and optoelectronics; photonic crystal fiber sensors; surface plasmon resonance-based biosensors; digital VLSI; signal processing and embedded systems.

RESEARCH EXPERIENCE

Undergraduate Thesis Research — Aug 2025 – Present
Supervisor: Prof. Mohammad Rakibul Islam, Department of EEE, IUT — Dhaka, Bangladesh
Thesis: *Quad-Sector Microstructure SPR-Based PCF Sensor with Enhanced DPSS*

- Designed a **quad-sector dual-plasmonic photonic crystal fiber (PCF) biosensor** incorporating Ag–TiO_2 and ITO layers to improve Double Peak Shift Sensitivity (DPSS) for refractive-index-based biosensing applications.
- Performed finite-element-method (FEM) simulations using **COMSOL Multiphysics 6.2** to characterize core–SPR mode coupling and investigate structural parameters, including air-hole diameter and pitch, for improved sensitivity and reduced confinement loss.
- Benchmarked the proposed design against **15+ recent PCF-SPR biosensors** reported in the literature, investigating dual-plasmonic layering as an approach for simultaneous wavelength- and amplitude-sensitivity enhancement.
- Achieved wavelength sensitivity, amplitude sensitivity, and DPSS values of up to **11,000 nm/RIU**, **141.5 RIU^{-1}**, and **9,000 nm/RIU**, respectively.

SELECTED ACADEMIC PROJECTS

Low-Power 6T Pass-Transistor XNOR Circuit — Jul 2026
VLSI Design | PSpice | Microwind

- Designed a low-power **6-transistor pass-transistor XNOR circuit** and evaluated its transient behavior using PSpice. Developed the corresponding stick diagram and Microwind layout for a compact digital-circuit implementation.
- Analyzed Power profiles, popagation delays and environmental impacts.

8051-Based Digital Alarm Clock Apr 2026
8051 Assembly | Proteus | Embedded Systems

- Implemented a software-based real-time clock in **8051 Assembly** using Timer 0 interrupts without an external RTC.
- Integrated three independent alarms, snooze functionality, 4×4 keypad input, LCD interfacing, and sequence-based alarm dismissal.

Deep Learning-Based Diabetic Foot Ulcer Detection Apr 2026
Deep Learning | Computer Vision | Python | TensorFlow/Keras

- Developed a computer-vision pipeline for diabetic foot ulcer classification using transfer-learning architectures including **MobileNetV2 and EfficientNet**, with image preprocessing and model evaluation.
- Incorporated **Grad-CAM** visualization for model interpretability and developed an interactive application for image-based prediction and visualization.

2-Port MIMO Microstrip Patch Antenna with Defected Ground Structure Mar 2026
Antenna Engineering | CST Studio Suite | RF/Microwave Design

- Independently designed and simulated a **two-port MIMO microstrip patch antenna** incorporating a defected ground structure (DGS) for improved inter-element isolation.
- Evaluated antenna characteristics including S_{11}, S_{21}, resonance behavior, radiation characteristics, gain, efficiency, and MIMO performance through full-wave electromagnetic simulation.

8086-Based Interactive Student Database Aug 2025
8086 Assembly | Microprocessor System Design

- Developed an interactive student database entirely in **8086 Assembly** with user-defined columns and data types, formatted table display, and DOS interrupt-based I/O.
- Implemented sorting of complete student records using **Bubble Sort** based on numeric or string fields.

Convolution and Correlation Visualizer Jul 2025
Digital Signal Processing | MATLAB | GUI Development

- Developed an interactive MATLAB application for visualizing **convolution and correlation** of discrete- and continuous-time signals.
- Implemented signal generation, transformation, graphical visualization, and step-by-step representation of fundamental signal-processing operations.

5-Bus Power-System Load Flow Analysis May 2024
Power Systems | MATLAB | PowerWorld Simulator

- Implemented **Gauss–Seidel load-flow analysis** for a five-bus power system consisting of slack, PV, and PQ buses.
- Computed bus voltages, phase angles, transmission-line power flows, and system losses, and compared MATLAB-based results with PowerWorld simulations.

6-Bit SAP-1 Computer Nov 2023
Digital Logic Design | Computer Architecture | Proteus

- Designed and integrated a **6-bit SAP-1 computer** around a common system bus with RAM, program counter, memory address register, registers, ALU, and controller-sequencer.
- Implemented `LDA`, `ADD`, `SUB`, `OUT`, and `HLT` instructions with manual and automatic clock operation.

TECHNICAL SKILLS

Research & Numerical Simulation: COMSOL Multiphysics, MATLAB, Simulink, CST Studio Suite, PowerWorld

Digital Design & EDA: Verilog HDL, SystemVerilog, RTL Design, combinational and sequential logic design, testbench development, functional verification, waveform analysis, RTL synthesis, Cadence Xcelium, Genus, Virtuoso, Microwind

Circuit & System Simulation: PSpice, OrCAD, Proteus, PSIM

Programming & Assembly: Python, C, MATLAB, 8051 Assembly, 8086 Assembly

Machine Learning & Data Analysis: TensorFlow/Keras, transfer learning, computer vision, model evaluation, Grad-CAM

Embedded Systems & Hardware: 8051 microcontroller, Arduino, sensor interfacing, PWM control, LCD/keypad interfacing, PCB design, hardware integration, testing, and troubleshooting

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Project Altair – IUT Mars Rover Team 2023 – 2025
Electrical System Sub-Team

- Served as **Senior Member, Electrical Sub-Team** during 2024–2025 after serving as Junior Member during 2023–2024.
- Contributed to circuit development, PCB design using Autodesk Eagle, wiring, system integration, testing, and troubleshooting of rover electrical systems.
- Participated in the **European Rover Challenge 2023** and **International Rover Challenge 2024** as part of Project Altair.
- Project Altair received the **Best Science Team Award at International Rover Challenge 2024**.

IUT Robotics Society 2023 – 2025

- Served as **Assistant Head (Event & Workshop)** during 2024–2025 and **Sub Executive Officer** during 2023–2024.
- Organized technical events including **TECHATHON 1.0 2024** and **Drone Workshop 2024**, contributing to planning, coordination, and execution.
- Contributed to the design of competition arenas for Soccerbot, Line Following Robot (LFR), and RC Bot competitions.

INDUSTRIAL TRAINING & VISITS

Ghorashal Training Center / Power Station Oct 2025
Industrial Training

- Gained exposure to power-station operations, electrical systems, power-generation processes, and associated industrial equipment.

Bangladesh Atomic Energy Commission (BAEC) Oct 2025
Industrial Training

- Observed solar-cell fabrication processes and visited the **3 MW TRIGA Mark-II research reactor**.

Robi Axiata Limited – Head Office Oct 2025
Industrial Visit

- Gained exposure to telecommunications operations, technical workflows, network infrastructure, and service management.

Walton Hi-Tech Industries PLC Oct 2025
Industrial Visit

- Observed industrial manufacturing processes, automation, production systems, and quality-control practices.

ADDITIONAL INFORMATION

Languages: Bangla (Native), English (Fluent)

REFERENCES

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